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Electrically driven hydraulic work machine

Abstract

In a hydraulic drive system used for an electrically driven hydraulic work machine that drives a hydraulic pump by an electric motor and supplies a hydraulic fluid to plural actuators to carry out work, the power consumption of the hydraulic pump is kept from exceeding a value decided in advance. For this purpose, a controller calculates target power to be consumed by the hydraulic pump on the basis of the capacity of the hydraulic pump, the delivery pressure of the hydraulic pump sensed by a pressure sensor, and a target revolution speed of the electric motor, and limits the target revolution speed of the electric motor in such a manner that the target power falls within a range of the maximum allowable power.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an electrically driven hydraulic work machine such as a hydraulic excavator that drives a hydraulic pump by an electric motor and carries out various kinds of work.

BACKGROUND ART

(2) Electrically driven hydraulic work machines such as hydraulic excavators that drive a hydraulic pump by an electric motor and carry out various kinds of work by plural actuators have been used in environments in which exhaust gas emission is unfavorable, for example in work environments of indoor site, underground, and so forth, because of characteristics such as a point that the electrically driven hydraulic work machines do not emit an exhaust gas due to an engine and a point that they have low noise.

(3) In patent document 1, an electrically driven hydraulic work machine is disclosed that includes, in addition to a built-in battery, a connector connected to a commercial power supply and a connector connected to an external battery, an AC-DC converter that converts AC power supplied from the connector connected to a commercial power supply to DC power and causes the DC power to merge into a line that supplies DC power from the built-in battery to an inverter for driving the electric motor, and a voltage adjuster that converts the voltage of DC power supplied from an external battery and causes the DC power to merge into the line that supplies DC power from the built-in battery to the inverter for driving the electric motor similarly to the above.

(4) When the technique of patent document 1 is used, since including the connector connected to a commercial power supply and the connector connected to an external battery, the electrically driven hydraulic work machine can drive a hydraulic pump by using commercial AC power fed through the connector connected to a commercial power supply or DC power supplied through the connector connected to an external battery even if the situation in which the remaining charge of the built-in battery is insufficient in operation occurs. This enables continuous operation of the electrically driven hydraulic work machine and it is possible to avoid the situation in which the electrically driven hydraulic work machine becomes inoperable at a construction site due to depletion of the built-in battery.

PRIOR ART DOCUMENT

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(6) However, also in patent document 1, there are the following problems.

(7) For example, when a work implement (for example, front work implement of a hydraulic excavator) is operated in the state in which the remaining charge of the built-in battery has become low, the battery voltage suddenly lowers due to the electric power consumption of the electric motor that drives the hydraulic pump and the battery voltage falls below the allowable range of the inverter that drives the electric motor, and thus the electrically driven hydraulic work machine suddenly stops, in some cases.

(8) Also in the case in which the electrically driven hydraulic work machine is operating by a commercial power supply through the connector connected to the commercial power supply, the electric power consumption (or current) of the electric motor that drives the hydraulic pump exceeds the power capacity (or current capacity) of the commercial power supply and a breaker included in the commercial power supply carries out interrupt operation, and thus the electrically driven hydraulic work machine suddenly stops operation and the work implement suddenly stops, in some cases.

(9) When the work implement of the electrically driven hydraulic work machine suddenly stops in operation as above, there are the case in which the stability of the work machine is impaired and a possibility of falling-down occurs, the case in which the inverter and the breaker need to be returned each time for example, that is, the convenience of the operator is impaired, and so forth.

(10) An object of the present invention is to provide an electrically driven hydraulic work machines that drive a hydraulic pump by an electric motor to carry out work, and in which the power consumption of the electric motor is prevented from exceeding a value decided in advance when the power to be consumed by the hydraulic pump increases and a sudden stop of a work implement that occurs due to abnormal lowering of the voltage of a built-in battery and operation of a breaker of a commercial power supply can be surely prevented.

Means for Solving the Problem

(11) In order to solve such a problem, the present invention provides an electrically driven hydraulic work machine comprising an electric motor, a hydraulic pump driven by the electric motor, and a controller that controls the revolution speed of the electric motor on the basis of a target revolution speed of the electric motor, and drives the hydraulic pump to carry out work, wherein the electrically driven hydraulic work machine further comprises a maximum allowable power setting device that sets maximum allowable power allowed to be consumed by the electric motor, and a pressure sensor that senses a delivery pressure of the hydraulic pump, and the controller is configured to calculate target power to be consumed by the hydraulic pump on the basis of the capacity of the hydraulic pump, the delivery pressure of the hydraulic pump sensed by the pressure sensor, and the target revolution speed of the electric motor and control the target revolution speed of the electric motor in such a manner that the target power falls within a range of the maximum allowable power.

Advantages of the Invention

(12) According to the present invention, the power consumed by the electric motor is surely limited to be equal to or lower than the maximum allowable power. This prevents abnormal lowering of the voltage of a built-in battery that supplies electric power to the electric motor and operation of a breaker of a commercial power supply to the interruption position in operation of the electrically driven hydraulic work machine and can surely prevent a sudden stop of a work implement.

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating the appearance of an electrically driven hydraulic work machine in a first embodiment.
- (2) FIG. 2 is a diagram illustrating a hydraulic drive system included in the electrically driven hydraulic work machine in the first embodiment.
- (3) FIG. 3 is a diagram illustrating the absorption torque characteristic of a main pump **2** controlled by a torque control piston **12d**.
- (4) FIG. 4 is a functional block diagram of a controller **50** in the first embodiment.
- (5) FIG. 5 is a diagram illustrating a hydraulic drive system included in an electrically driven hydraulic work machine of a second embodiment.
- (6) FIG. 6 is a diagram illustrating the absorption torque characteristic of a main pump controlled by torque control pistons.
- (7) FIG. 7 is a diagram illustrating the absorption torque characteristic of a main pump of the fixed displacement type.
- (8) FIG. 8 is a functional block diagram of a controller **55** in the second embodiment.

MODES FOR CARRYING OUT THE INVENTION

- (9) Embodiments of the present invention will be described below according to the drawings.

First Embodiment

- (10) —Configuration—

- (11) FIG. 1 is a diagram illustrating the appearance of an electrically driven hydraulic work machine in a first embodiment of the present invention.
- (12) The electrically driven hydraulic work machine includes a lower track structure **101**, an upper swing structure **102**, and a front work implement **104** of a swing type, and the front work implement **104** is composed of a boom **111**, an arm **112**, and a bucket **113**. The upper swing structure **102** and the lower track structure **101** are rotatably connected to each other by a swing ring **215** and the upper swing structure **102** can swing relative to the lower track structure **101** by rotation of a swing motor **3c**. A swing post **103** is attached to the front part of the upper swing structure **102** and the front work implement **104** is attached to this swing post **103** vertically movably. The swing post **103** can be pivoted in the horizontal direction relative to the upper swing structure **102** by extension and contraction of a swing cylinder **3e**. The boom **111**, the arm **112**, and the bucket **113** of the front work implement **104** can be pivoted in the upward-downward direction by extension and contraction of a boom cylinder **3a**, an arm cylinder **3b**, and a bucket cylinder **3d**. To a center frame of the lower track structure **101**, right and left track devices **105a** and **105b** and a blade **106** that carries out upward-downward operation by extension and contraction of a blade cylinder **3h** are attached. The right and left track devices **105a** and **105b** include drive wheels **210a** and **210b**, idlers **211a** and **211b**, and crawlers **212a** and **212b**, respectively, and carry out travelling by driving the crawlers **212a** and **212b** through the drive wheels **210a** and **210b** by rotation of right and left travelling motors **3f** and **3g**.
- (13) In the upper swing structure **102**, a battery mounting part **109** in which a battery **70** is mounted on a swing frame **107** and a cabin **110** inside which an operation room **108** is formed are set. In the operation room **108**, an operation seat **122**, right and left operation lever devices **124A** and **124B** for the boom cylinder **3a**, the arm cylinder **3b**, the bucket cylinder **3d**, and the swing motor **3c**, a monitor **80**, and a gate lock lever **24** (see FIG. 2) are disposed.
- (14) FIG. 2 is a diagram illustrating a hydraulic drive system included in the electrically driven hydraulic work machine in the first embodiment.
- (15) The hydraulic drive system includes an electric motor **1**, a main hydraulic pump (hereinafter, referred to as main pump) **2** of the variable displacement type driven by the electric motor **1**, and a pilot pump **30** of the fixed displacement type. The hydraulic drive system includes also the boom cylinder **3a**, the arm cylinder **3b**, the swing motor **3c**, the bucket cylinder **3d** (see FIG. 1), the swing

cylinder **3e** (see FIG. 1), the travelling motors **3f** and **3g** (see FIG. 1), and the blade cylinder **3h** (see FIG. 1) that are plural actuators driven by a hydraulic fluid delivered from the main pump **2**. The hydraulic drive system includes also a hydraulic fluid supply line **5** for introducing the hydraulic fluid delivered from the main pump **2** to the plural actuators **3a, 3b, 3c, 3d, 3e, 3f, 3g**, and **3h** and a control valve block **4** that is connected to the downstream side of the hydraulic fluid supply line **5** and to which the hydraulic fluid delivered from the main pump **2** is introduced. Hereinafter, the “actuators **3a, 3b, 3c, 3d, 3e, 3f, 3g**, and **3h**” will be represented with simplification to the “actuators **3a, 3b, 3c . . .**”

(16) The control valve block **4** is a control valve device that distributes and supplies the hydraulic fluid delivered from the main pump **2** to the plural actuators **3a, 3b, 3c . . .**. In the control valve block **4**, plural directional control valves **6a, 6b, 6c . . .** for controlling the plural actuators **3a, 3b, 3c . . .** and plural pressure compensating valves **7a, 7b, 7c . . .** each located on the downstream side of the respective meter-in openings of the plural directional control valves **6a, 6b, 6c . . .** are disposed. To the plural pressure compensating valves **7a, 7b, 7c . . .**, the pressure of the upstream side of the meter-in openings of the directional control valves **6a, 6b, 6c . . .** is introduced in such a direction as to bias a spool of the pressure compensating valves **7a, 7b, 7c . . .** in the closing direction and the load pressure of the actuators **3a, 3b, 3c . . .** and the output pressure of a differential pressure reducing valve **11** to be described later are introduced in such a direction as to bias the spool in the opening direction. Between the pressure compensating valves **7a, 7b, 7c . . .** and the directional control valves **6a, 6b, 6c . . .**, check valves **8a, 8b, 8c . . .** that each prevent backflow of the hydraulic fluid from the directional control valve **6a, 6b, 6c . . .** to the pressure compensating valve **7a, 7b, 7c . . .** are disposed.

(17) Furthermore, in the control valve block **4**, shuttle valves **9a, 9b, 9c . . .** connected to load pressure sensing ports of the plural directional control valve **6a, 6b, 6c . . .** are disposed. The shuttle valves **9a, 9b, 9c . . .** are connected into a tournament format and the highest load pressure is sensed by the shuttle valve **9a** of the highest level and is output to a hydraulic line **8**.

(18) Moreover, in the control valve block **4**, on the downstream side of the hydraulic fluid supply line **5**, a main relief valve **14** that discharges the hydraulic fluid of the hydraulic fluid supply line **5** to a tank when the pressure of the hydraulic fluid supply line **5** (delivery pressure of the main pump **2**) has become equal to or higher than a set pressure decided in advance, the differential pressure reducing valve **11** that outputs the differential pressure between the pressure (delivery pressure of the main pump **2**) P_{ps} of the hydraulic fluid supply line **5** and a highest load pressure P_{plmax} to be described later as an absolute pressure $P_{ls} (=P_{ps}-P_{plmax})$, and an unloading valve **15** that discharges the hydraulic fluid of the hydraulic fluid supply line **5** to the tank when the differential pressure between the pressure (delivery pressure of the main pump **2**) P_{ps} of the hydraulic fluid supply line **5** and the highest load pressure P_{plmax} has become equal to or higher than a certain set pressure (unloading differential pressure) are disposed. The unloading valve **15** has pressure receiving parts **15a** and **15d** and a spring **15b** that bias a spool of the unloading valve **15** in the closing direction and a pressure receiving part **15c** that biases the spool in the opening direction. The highest load pressure P_{plmax} of the plural actuators **3a, 3b, 3c . . .** is introduced to the pressure receiving part **15a**. An output pressure P_{gr} (target LS differential pressure) of a prime mover revolution speed sensing valve **13** to be described later is introduced to the pressure receiving part **15d**. The pressure (delivery pressure of the main pump **2**) P_{ps} of the hydraulic fluid supply line **5** is introduced to the pressure receiving part **15c**. The unloading differential pressure of the unloading valve **15** is set based on the spring constant of the spring **15b** and the output pressure of the prime mover revolution speed sensing valve **13** (target LS differential pressure P_{gr}) introduced to the pressure receiving part **15d**.

(19) The main pump **2** of the variable displacement type has a regulator **12** and the regulator **12** includes a torque control piston **12d** to which the pressure (delivery pressure of the main pump **2**) P_{ps} of the hydraulic fluid supply line **5** is introduced and that controls the capacity (tilting angle) of

the main pump **2** in such a manner that the absorption torque of the main pump **2** does not exceed a predetermined value set based on a spring **12e**.

(20) FIG. **3** is a diagram illustrating the absorption torque characteristic of the main pump **2** controlled by the torque control piston **12d**. In FIG. **3**, the abscissa axis is the delivery pressure P_{ps} of the main pump **2** and the ordinate axis is capacity q (tilting angle) of the main pump **2**.

(21) Until the delivery pressure P_{ps} of the main pump **2** rises to P_{pq1} , the capacity q of the main pump **2** is equal to a maximum capacity q_{max} that depends on specifications of the main pump **2**. When the delivery pressure P_{ps} has risen to P_{pq1} or higher, the capacity q gradually becomes lower from the maximum capacity q_{max} as the delivery pressure P_{ps} rises. When the delivery pressure P_{ps} reaches P_{pq2} , the capacity q becomes equal to q_{min} . While the delivery pressure is in the range from P_{pq1} to P_{pq2} , the absorption torque of the main pump **2** is kept at the predetermined value set based on the spring **12e**. P_{pq2} is the maximum pressure that depends on the set pressure of the main relief valve **14**.

(22) Furthermore, the regulator **12** includes a flow rate control piston **12c** that controls the delivery flow rate of the main pump **2** and an LS valve **12b** that switches whether to introduce a constant pilot pressure P_{i0} generated by a pilot relief valve **32** to be described later to the flow rate control piston **12c** or to discharge the pressure of the flow rate control piston **12c** to the tank.

(23) To the LS valve **12b**, the output pressure P_{ls} of the differential pressure reducing valve **11** is introduced in such a direction as to carry out switching to introduce the constant pilot pressure P_{i0} to the flow rate control piston **12c** and the output pressure P_{gr} (target LS differential pressure) of the prime mover revolution speed sensing valve **13** is introduced in such a direction as to carry out switching to discharge the hydraulic fluid of the flow rate control piston **12c** to the tank. The LS valve **12b** and the flow rate control piston **12c** control the capacity of the main pump **2** in such a manner that the pressure (delivery pressure of the main pump **2**) P_{ps} of the hydraulic fluid supply line **5** becomes higher than the highest load pressure P_{lmax} of the actuator driven by the hydraulic fluid delivered from the main pump **2** by the output pressure P_{gr} (target LS differential pressure) of the prime mover revolution speed sensing valve **13**.

(24) The prime mover revolution speed sensing valve **13** is disposed on a pilot pressure supply line **31a** of the pilot pump **30** and senses the revolution speed of the electric motor **1** from the delivery flow rate of the pilot pump **30**. The prime mover revolution speed sensing valve **13** has a flow rate sensing valve **13a** connected between the hydraulic fluid supply line **31a** of the pilot pump **30** and a pilot hydraulic pressure supply line **31b** and a differential pressure reducing valve **13b** that outputs the differential pressure across the flow rate sensing valve **13a** as the target LS differential pressure P_{gr} . On the pilot pressure supply line **31b** on the downstream side of the prime mover revolution speed sensing valve **13**, the pilot relief valve **32** that keeps the pressure of the pilot pressure supply line **31b** constant and forms a pilot hydraulic fluid source on the pilot pressure supply line **31b** and a selector valve **100** that switches whether or not to supply the pressure of the pilot pressure supply line **31b** to plural pilot valves (pressure reducing valves) that are for actuating the plural directional control valves **6a**, **6b**, **6c** . . . and are not illustrated in the diagram are disposed. The plural pilot valves are each incorporated in plural operation lever devices including the operation lever devices **124A** and **124B** (see FIG. **1**) for the boom cylinder **3a**, the arm cylinder **3b**, the bucket cylinder **3d**, and the swing motor **3c**, and are actuated by operating the operation lever of the corresponding operation lever device to generate an operation pilot pressure for actuation of the plural directional control valves **6a**, **6b**, **6c** . . . with use of the hydraulic fluid introduced from the pilot pressure supply line **31b** via a pilot pressure supply line **31c** as a pilot primary pressure.

(25) For the selector valve **100**, the above-described gate lock lever **24** for switching whether or not to permit operation of the operation lever of the operation lever device is disposed. In the selector valve **100**, through operation of the gate lock lever **24** by the operator in the operation room **108** (see FIG. **1**), whether the pressure of the pilot pressure supply line **31b** is supplied to the plural pilot valves (not illustrated) as the pilot primary pressure or the pilot primary pressure supplied to

the pilot valves is discharged to the tank is switched.

(26) Next, characteristic configurations of the electrically driven hydraulic work machine in the present embodiment will be described.

(27) In the present embodiment, the main pump **2** is a hydraulic pump driven by the electric motor **1** and the electrically driven hydraulic work machine is an electrically driven hydraulic work machine that drives the main pump **2** to carry out work. Furthermore, the electrically driven hydraulic work machine includes a controller **50** that controls the revolution speed of the electric motor **1** on the basis of a target revolution speed of the electric motor **1**. The controller **50** calculates target power to be consumed by the main pump **2** on the basis of the capacity of the main pump **2** (hydraulic pump), the delivery pressure of the main pump **2** sensed by a pressure sensor **41**, and the target revolution speed set in advance regarding the electric motor **1**, and limits the target revolution speed of the electric motor **1** in such a manner that the target power falls within a range of the maximum allowable power. Details thereof will be described below.

(28) In the present embodiment, the hydraulic drive system includes an inverter **60** for controlling the revolution speed of the electric motor **1** and a battery **70** connected so as to supply DC power to the inverter **60** through a DC power supply line **65**. Furthermore, the hydraulic drive system includes an AC/DC converter **90** connected to the DC power supply line **65** and a connector **91** connected to the AC/DC converter **90** and is configured to allow DC power to be supplied to the inverter **60** through the connector **91** and the AC/DC converter **90** on the basis of AC power supplied from a commercial power supply **92** when the commercial power supply **92** is connected to the connector **91**.

(29) Moreover, the hydraulic drive system includes a target revolution speed instruction dial (target revolution speed instruction device) **51** to make an instruction of the target revolution speed of the electric motor **1**, a monitor **80** in which a maximum allowable power setting device **81** that sets the maximum allowable power that can be consumed by the electric motor **1** is incorporated, and the pressure sensor **41** that is connected to the hydraulic fluid supply line **5** and senses the pressure of the hydraulic fluid supply line **5** as the delivery pressure P_{ps} of the main pump **2**. The output of the pressure sensor **41**, the output of the target revolution speed instruction dial **51**, and the output of the maximum allowable power setting device **81** are each introduced to the controller **50**. The controller **50** outputs the target revolution speed of the electric motor **1** to the inverter **60** as a command revolution speed.

(30) In the maximum allowable power setting device **81** incorporated in the monitor **80**, plural values of the maximum allowable power corresponding to the power supply that supplies electric power to the electric motor **1** are stored according to the kinds of power supply. The maximum allowable power setting device **81** is configured to select what corresponds to the battery **70** and the commercial power supply **92** that are the power supplies that supply electric power to the electric motor **1** from the stored values of the maximum allowable power and set the maximum allowable power. For example, current values are stored as the maximum allowable power.

(31) FIG. **4** is a functional block diagram of the controller **50** in the first embodiment.

(32) In FIG. **4**, the controller **50** has, as processing functions thereof, a table **50a**, a multiplying section **50b**, a multiplying section **50c**, a minimum value selecting section **50d**, a dividing section **50e**, a dividing section **50f**, and a minimum value selecting section **50g**.

(33) In the table **50a**, the same characteristic as the absorption torque characteristic (see FIG. **3**) of the main pump **2** controlled by the torque control piston **12d** of the above-described regulator **12** is set. The delivery pressure P_{ps} of the main pump **2** that is an output from the pressure sensor **41** is introduced to the table **50a**. Reference to the delivery pressure P_{ps} of the main pump **2** is made in the table **50a** and the capacity q of the main pump **2** is calculated.

(34) The main pump **2** may be the fixed displacement type. In this case, like a hydraulic pump **21** in a second embodiment to be described later, a table in which constant capacity q_{max} like one illustrated in FIG. **7** is set is prepared, and the capacity can be calculated from the delivery pressure

of the main pump at the time. Furthermore, the constant capacity $q_{\max}$ may be stored in a memory of the controller **50** and the capacity $q_{\max}$ may be used.

(35) A target revolution speed N_{ac} that is an input from the target revolution speed instruction dial **51** is introduced to the multiplying section **50b** together with the capacity q calculated with the table **50a** and a target flow rate Q_{ac} is calculated. This target flow rate Q_{ac} and the delivery pressure P_{ps} of the main pump **2** that is the output from the pressure sensor **41** are introduced to the multiplying section **50c** and target power P_{wac} is calculated.

(36) Moreover, maximum allowable power $P_{w\max}$ that is an output from the maximum allowable power setting device **81** incorporated in the monitor **80** and the target power P_{wac} calculated in the multiplying section **50c** are introduced to the minimum value selecting section **50d** and post-limiting power P_{wreg} is calculated. The post-limiting power P_{wreg} and the delivery pressure P_{ps} of the main pump **2** that is the output from the pressure sensor **41** are introduced to the dividing section **50e** and a post-limiting flow rate Q_{reg} is calculated. The post-limiting flow rate Q_{reg} and the capacity q calculated with the table **50a** are introduced to the dividing section **50f** and a post-limiting revolution speed N_{reg} is calculated.

(37) The post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} that is the input from the target revolution speed instruction dial **51** are input to the minimum value selecting section **50g** and the smaller value of the post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} is selected as a command revolution speed N_d and is output to the inverter **60**.

(38) As above, the controller **50** calculates the first target revolution speed (post-limiting revolution speed) N_{reg} of the electric motor **1** on the basis of the post-limiting power P_{wreg} that is the lower power of the target power P_{wac} and the maximum allowable power $P_{w\max}$ set by the maximum allowable power setting device **81**, and selects the lower target revolution speed of this first target revolution speed N_{reg} and the target revolution speed N_{ac} of the electric motor **1** regarding which an instruction is made by the target revolution speed instruction device (target revolution speed instruction dial) **51** as the second target revolution speed (command target revolution speed) N_d and controls the revolution speed of the electric motor **1** on the basis of the second target revolution speed N_d .

(39) —Operation—

(40) Operation of the first embodiment will be described.

(41) The hydraulic fluid delivered from the pilot pump **30** of the fixed displacement type is supplied to the pilot pressure supply line **31a** and the prime mover revolution speed sensing valve **13** outputs the target LS differential pressure P_{gr} according to the delivery flow rate of the pilot pump **30**. The pilot primary pressure generated by the pilot relief valve **32** is supplied to the respective pilot valves of the plural operation lever devices including the operation lever devices **124A** and **124B** through the selector valve **100** actuated to be switched by the gate lock lever.

(42) When the operation lever of an optional operation lever device in the plural operation lever devices including the operation lever devices **24A** and **124B** (see FIG. 1) is operated, the corresponding pilot valve is actuated and the corresponding directional control valve is switched, and thus the hydraulic fluid is supplied to the corresponding actuator. At this time, the directional control valve is switched with a stroke according to the operation amount of the operation lever and the main pump **2** driven by the electric motor **1** delivers a flow rate according to the operation amount of the operation lever on the basis of load sensing control by the LS valve **12b** and the flow rate control piston **12c** of the regulator **12**. The actuator is driven at a speed according to the operation amount of the operation lever.

(43) In the present embodiment, the flow rate control of the main pump **2** by the LS valve **12b** and the flow rate control piston **12c** is general load sensing control. Therefore, details thereof are omitted.

(44) DC power supplied from the battery **70** or DC power supplied after being converted from AC

power by the AC/DC converter **90** from the commercial power supply **92** through the connector **91** or the DC power of both is supplied to the inverter **60** that drives the electric motor **1** through the DC power supply line **65**.

(45) The maximum allowable power P_{wmax} set in advance is input to the controller **50** from the maximum allowable power setting device **81** incorporated in the monitor **80**.

(46) The output from the pressure sensor **41** as the pump delivery pressure P_{ps} and the output from the target revolution speed instruction dial **51** as the target revolution speed N_{ac} are each input to the controller **50**.

(47) Processing in the controller **50** will be described below with classification of cases.

(48) (a) Case in which the Target Power P_{wac} of the Main Pump **2** is the Same as or Lower than the Maximum Allowable Power P_{wmax} ($P_{wac} \leq P_{wmax}$)

(49) The maximum allowable power P_{wmax} and the target power P_{wac} are introduced to the minimum value selecting section **50d** and P_{wac} that is the minimum value is selected, and thus the post-limiting power P_{wreg} becomes $P_{wreg} = P_{wac}$.

(50) In the dividing section **50e**, P_{wreg}/P_{ps} is computed. At this time, $P_{wreg} = P_{wac}$ holds in the case of $P_{wac} \leq P_{wmax}$. Therefore, the post-limiting flow rate Q_{reg} becomes $Q_{reg} = P_{wreg}/P_{ps} = P_{wac}/P_{ps} = Q_{ac}$.

(51) In the dividing section **50f**, Q_{reg}/q is computed. At this time, $Q_{reg} = Q_{ac}$ holds as described above. Therefore, the post-limiting revolution speed N_{reg} becomes $N_{reg} = Q_{reg}/q = Q_{ac}/q = N_{ac}$.

(52) The post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} are input to the minimum value selecting section **50g** and the minimum value is selected. At this time, $N_{reg} = N_{ac}$ holds as described above. Therefore, the command revolution speed N_d output from the controller **50** to the inverter **60** becomes $N_d = N_{ac}$ without being limited by the minimum value selecting section **50g**.

(53) (b) Case in which the Target Power P_{wac} of the Main Pump **2** is Higher than the Maximum Allowable Power P_{wmax} ($P_{wac} > P_{wmax}$)

(54) The maximum allowable power P_{wmax} and the target power P_{wac} are each introduced to the minimum value selecting section **50d**. In this case, the maximum allowable power P_{wmax} is selected as the minimum value and the post-limiting power P_{wreg} becomes $P_{wreg} = P_{wmax}$.

(55) The post-limiting flow rate Q_{reg} is calculated as $Q_{reg} = P_{wmax}/P_{ps}$ by the dividing section **50e**. At this time, since originally a relation of $Q_{ac} = P_{wac}/P_{ps}$ holds, a relation of $Q_{reg}/Q_{ac} = P_{wmax}/P_{wac} (<1)$ holds from these two expressions.

(56) Subsequently, the post-limiting revolution speed N_{reg} is calculated as $N_{reg} = Q_{reg}/q = P_{wmax}/P_{ps}/q$ by the dividing section **50f**. Also in this case, since originally a relation of $N_{ac} = Q_{ac}/q$ holds, a relation of $N_{reg}/N_{ac} = Q_{reg}/Q_{ac} = P_{wmax}/P_{wac} (<1)$ holds from these two expressions.

(57) The post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} are input to the minimum value selecting section **50g**. At this time, since $N_{reg} < N_{ac}$ holds as described above, N_{reg} that is the smaller value than the target revolution speed N_{ac} is selected as the command revolution speed N_d and is output from the controller **50** to the inverter **60**.

(58) —Effects—

(59) In the present embodiment, the following effects are obtained.

(60) 1. The controller **50** calculates the target power P_{wac} to be consumed by the main pump **2** on the basis of the capacity q of the main pump **2**, the delivery pressure P_{ps} of the main pump **2** sensed by the pressure sensor **41**, and the target revolution speed N_{ac} of the electric motor **1** and outputs the command revolution speed N_d to the inverter **60** to limit the target revolution speed N_{ac} of the electric motor **1** in such a manner that the target power P_{wac} falls within the range of the maximum allowable power P_{wmax} . Therefore, the power consumption of the electric motor **1** is surely limited to be equal to or lower than the maximum allowable power P_{wmax} . This prevents abnormal lowering of the voltage of the battery **70** that supplies electric power to the electric motor

1 and operation of the breaker of the commercial power supply **92** to the interruption position in operation of the electrically driven hydraulic work machine and can surely prevent a sudden stop of the front work implement **104**.

(61) 2. Furthermore, without manual operation of the target revolution speed instruction dial **51** by the operator, the target revolution speed N_{ac} of the electric motor **1** is limited in such a manner that the power consumption of the electric motor **1** does not exceed the maximum allowable power P_{wmax} . Therefore, a sudden stop of the front work implement **104** that occurs due to abnormal lowering of the voltage of the battery **70** and operation of the breaker of the commercial power supply **92** is surely prevented. In addition, the lowering of the work efficiency can be suppressed to the minimum without lowering the operation speed of the front work implement **104** beyond necessity.

(62) Specifically, in general, it is known that the electric power consumed by the electric motor of the electrically driven hydraulic work machine is almost equal to the power consumption of the hydraulic pump driven by the electric motor and is proportional to “delivery pressure”×“delivery flow rate,” and the delivery flow rate is proportional to the revolution speed of the electric motor. Therefore, when the electric power consumption of the electric motor is desired to be suppressed, setting the instruction value of the target revolution speed instruction dial small by the operator is generally carried out. However, the operator oneself needs to learn how small the instruction value of the target revolution speed instruction dial is set in order to allow prevention of a stop of the electrically driven hydraulic work machine in operation while carrying out actual work. This is a cause of impairing the comfort of the operator due to troublesomeness thereof. Furthermore, when the instruction value of the target revolution speed instruction dial is set too small, the load of the hydraulic pump of the electrically driven hydraulic work machine is low and the operation speed of the work implement becomes low even when the revolution speed does not need to be suppressed to a low speed. This is a cause of lowering the work efficiency.

(63) In the present embodiment, the controller **50** calculates the target power P_{wac} to be consumed by the main pump **2** on the basis of the capacity q of the main pump **2**, the delivery pressure P_{ps} of the main pump **2** sensed by the pressure sensor **41**, and the target revolution speed N_{ac} of the electric motor **1** regarding which an instruction is made by the target revolution speed instruction dial **51**. Thus, the operator of the electrically driven hydraulic work machine does not need to operate the target revolution speed instruction dial **51** of the electric motor **1** to limit the revolution speed of the electric motor **1**, and troublesomeness of operation can be eliminated.

(64) Furthermore, the situation does not occur in which, when the target power of the electric motor **1** is low, the revolution speed of the electric motor **1** is unnecessarily limited and the operation speed of the front work implement **104** is lowered. Therefore, the lowering of the work efficiency of the electrically driven hydraulic work machine can be suppressed to the minimum.

(65) 3. The maximum allowable power setting device **81** is configured to select what corresponds to the battery **70** and the commercial power supply **92** that are power supplies that supply electric power to the electric motor **1** from plural values of the maximum allowable power stored in advance and set the maximum allowable power. Therefore, even an operator who is inexperienced in handling of the electrically driven hydraulic work machine can easily set the maximum allowable power.

(66) 4. The controller **50** sets the same characteristic as the absorption torque characteristic (see FIG. **4**) of the main pump **2** in the table **50a** and refers to the delivery pressure P_{ps} of the main pump **2** sensed by the pressure sensor **1** in the table **50a** to calculate the capacity q of the main pump **2**. Therefore, it is possible to accurately calculate the absorption torque of the main pump **2** and surely prevent abnormal lowering of the voltage of the battery **70** that supplies electric power to the electric motor **1** and operation of the breaker of the commercial power supply **92** to the interruption position.

(67) 5. The controller **50** does not output, to the inverter **60**, the post-limiting revolution speed (first

target revolution speed) Nreg of the electric motor **1** calculated based on the post-limiting power Pwreg as the command revolution speed Nd as it is but outputs, to the inverter **60**, the lower target revolution speed (second target revolution speed) of the post-limiting revolution speed Nreg and the target revolution speed Nac regarding which an instruction is made by the target revolution speed instruction dial **51** as the command revolution speed Nd to control the revolution speed of the electric motor **1**. Therefore, when the target power Pwac of the main pump **2** is the same as or lower than the maximum allowable power Pwmax ($Pwac \leq Pwmax$), stable revolution speed control of the electric motor **1** can be carried out without being affected by the processing speed or the responsiveness of the controller **50**.

Second Embodiment

(68) Regarding the second embodiment of the present invention, configuration, operation, and effects thereof will be described with focus on the different part from the first embodiment.

(69) —Configuration—

(70) FIG. **5** is a diagram illustrating a hydraulic drive system included in an electrically driven hydraulic work machine of the second embodiment.

(71) In the second embodiment, the hydraulic drive system is different from the first embodiment in that the hydraulic pump is a hydraulic pump for which flow rate control based on load sensing is not carried out and two hydraulic pumps (first and second hydraulic pumps) are included as the hydraulic pump, in that one of the two hydraulic pumps is the split-flow type and, corresponding thereto, three pressure sensors are included as pressure sensors that sense the delivery pressure of the hydraulic pumps, in that a control valve block is a control valve device including directional control valves of the open center type that do not carry out flow dividing control, and in that a regulator of the hydraulic pumps is configured to carry out total torque control (torque control in which, when plural hydraulic pumps exist, the capacity of one hydraulic pump is controlled in such a manner that the total of the absorption torque of the plural hydraulic pumps does not exceed a predetermined value).

(72) In FIG. **5**, the hydraulic drive system of the present embodiment includes a main hydraulic pump (first hydraulic pump) **20** of the variable displacement type that is a hydraulic pump of the split-flow type driven by the electric motor **1** and a main hydraulic pump (second hydraulic pump) **21** that is a hydraulic pump of the fixed displacement type. The main pump **20** of the split-flow type has two delivery ports **20a** and **20b** that deliver a hydraulic fluid pushed out from a common pumping mechanism including swash plate, piston, and so forth and the hydraulic fluid delivered from the delivery ports **20a** and **20b** is supplied to the respective directional control valves.

(73) The main pump **20** may be a hydraulic pump having one delivery port. Furthermore, the main pump **20** may be two or more hydraulic pumps having one delivery port.

(74) Moreover, the hydraulic drive system of the present embodiment includes a hydraulic fluid supply line **5a** for introducing the hydraulic fluid delivered from one delivery port **20a** of the main pump **20** to plural actuators **3a**, **3d**, and **3g**, a hydraulic fluid supply line **5b** for introducing the hydraulic fluid delivered from the other delivery port **20b** of the main pump **20** to plural actuators **3b** and **3f**, a hydraulic fluid supply line **5c** for introducing the hydraulic fluid delivered from the main pump **21** to plural actuators **3c**, **3e**, and **3h**, and a control valve block **40** that is connected to the downstream side of the hydraulic fluid supply lines **5a**, **5b**, and **5c** and to which the hydraulic fluid delivered from the main pumps **20** and **21** is introduced. The plural actuators **3a**, **3b**, **3c**, **3d**, **3e**, **3f**, **3g**, and **3h** are boom cylinder, arm cylinder, swing motor, bucket cylinder, swing cylinder, travelling motors, and blade cylinder, respectively, as described in the first embodiment.

(75) The control valve block **40** is a control valve device that distributes and supplies the hydraulic fluid delivered from the main pumps **20** and **21** to the plural actuators **3a**, **3b**, **3c**, **3d**, **3e**, **3f**, **3g**, and **3h**. In the control valve block **40**, plural directional control valves **16a**, **16b**, **16c**, **16d**, **16e**, **16f**, **16g**, and **16h** for controlling the plural actuators **3a**, **3b**, **3c**, **3d**, **3e**, **3f**, **3g**, and **3h** and main relief valves **14a**, **14b**, and **14c** that are connected to the hydraulic fluid supply lines **5a**, **5b**, and **5c** and

discharge the hydraulic fluid of the hydraulic fluid supply lines **5a**, **5b**, and **5c** to a tank when the pressure of the hydraulic fluid supply lines **5a**, **5b**, and **5c** has become equal to or higher than a set pressure decided in advance are disposed. Between the hydraulic fluid supply lines **5a**, **5b**, and **5c** and the plural directional control valves **16a**, **16b**, **16c**, **16d**, **16e**, **16f**, **16g**, and **16h**, check valves **18a**, **18b**, **18c**, **18d**, **18e**, **18f**, **18g**, and **18h** that each prevent backflow of the hydraulic fluid from the directional control valve **16a**, **16b**, **16c**, **16d**, **16e**, **16f**, **16g**, or **16h** to the hydraulic fluid supply line **5a**, **5b**, or **5c** are disposed.

(76) The main pump **20** of the variable displacement type has a regulator **22**. The regulator **22** includes torque control pistons **22f**, **22g**, and **22h** to which the pressures of the hydraulic fluid supply lines **5a** and **5b** (delivery pressures of the two delivery ports **20a** and **20b** of the main pump **20**) and the hydraulic fluid of the hydraulic fluid supply line **5c** are each introduced and that control the capacity (tilting angle) of the main pump **20** in such a manner that the total of the absorption torque of the main pump **20** and the absorption torque of the main pump **21** does not exceed a predetermined value set based on a spring **22e**.

(77) FIG. **6** is a diagram illustrating the absorption torque characteristic of the main pump **20** controlled by the torque control pistons **22f**, **22g**, and **22h**. In FIG. **6**, the abscissa axis is an average delivery pressure $(Pps1+Pps2)/2$ of the main pump **20** and the ordinate axis is capacity $q12$ (tilting angle) of the main pump **20**. Furthermore, $Pps1$ and $Pps2$ are the delivery pressures of the two delivery ports **20a** and **20b**, respectively, of the main pump **20** and $Pps3$ is the delivery pressure of the main pump **21**.

(78) The capacity $q12$ of the main pump **20**, similarly to the absorption torque characteristic of the regulator **12** in the first embodiment illustrated in FIG. **3**, is equal to a maximum capacity $qmax12$ that depends on specifications of the main pump **20** until the average delivery pressure $(Pps1+Pps2)/2$ of the main pump **20** rises to $Ppq1a-Ppq1c$. When the average delivery pressure $(Pps1+Pps2)/2$ has risen to $Ppq1a-Ppq1c$ or higher, the capacity $q12$ gradually becomes lower from the maximum capacity $qmax12$ as the average delivery pressure $(Pps1+Pps2)/2$ rises. When the average delivery pressure $(Pps1+Pps2)/2$ reaches $Ppq2$, the capacity $q12$ becomes equal to $qmin12a-qmin12c$. While the average delivery pressure $(Pps1+Pps2)/2$ is in the range of $Ppq1a-Ppq1c$ to $Ppq2$, the absorption torque of the main pump **20** is kept at the predetermined value set based on the spring **22e**. $Ppq2$ is the maximum pressure that depends on the set pressure of the main relief valves **14a** and **14b**.

(79) The characteristic between $Ppq1a-Ppq1c$ to $Ppq2$ changes depending on the magnitude of the delivery pressure $Pps3$ of the main pump **21**. The characteristic becomes a characteristic on a curve a when the value of the delivery pressure $Pps3$ is small, becomes a characteristic on a curve c when the value of the delivery pressure $Pps3$ is large, and becomes a characteristic on a curve b when the value of the delivery pressure $Pps3$ is in the middle.

(80) The pilot pump **30** of the fixed displacement type is directly connected to the pilot pressure supply line **31b** and the pilot relief valve **32** and the selector valve **100** are disposed on the pilot pressure supply line **31b** similarly to the first embodiment.

(81) FIG. **7** is a diagram illustrating the absorption torque characteristic of the main pump **21** of the fixed displacement type. In FIG. **7**, the abscissa axis is the delivery pressure $Pps3$ of the main pump **21** and the ordinate axis is capacity $q3$ (tilting angle) of the main pump **21**. Since the main pump **21** is the fixed displacement type, the capacity is constant at $qmax3$ irrespective of the value of the delivery pressure $Pps3$ of the main pump **21**. $Ppq3$ is the maximum pressure that depends on the set pressure of the main relief valve **14c**.

(82) Furthermore, in the present embodiment, the hydraulic drive system includes a controller **55** that outputs a target revolution speed of the electric motor **1** to the inverter **60** as a command revolution speed, and includes pressure sensors **41a** and **41b** that are connected to the hydraulic fluid supply lines **5a** and **5b** and sense the delivery pressures $Pps1$ and $Pps2$ of the two delivery ports **20a** and **20b** of the main pump **20** and a pressure sensor **41c** that is connected to the hydraulic

fluid supply line **5c** and senses the delivery pressure **Pps3** of the main pump **21**. The outputs of the pressure sensors **41a**, **41b**, and **41c**, the output of the target revolution speed instruction dial **51**, and the output of the maximum allowable power setting device **81** are each introduced to the controller **55**.

(83) FIG. **8** is a functional block diagram of the controller **55** in the second embodiment.

(84) In FIG. **8**, the controller **55** has, as processing functions thereof, an adding section **55a**, a gain **55b**, a table **55c**, a gain **55d**, a multiplying section **55e**, a multiplying section **55f**, an adding section **55g**, a table **55h**, a multiplying section **55i**, a multiplying section **55j**, a minimum value selecting section **55k**, a gain **55l**, a dividing section **55m**, a dividing section **55n**, and a minimum value selecting section **55o**.

(85) The delivery pressures **Pps1** and **Pps2** of the main pump **20** that are outputs from the pressure sensors **41a** and **41b** are introduced to the adding section **55a** and the sum is turned to $\frac{1}{2}$ by the gain **55b**, and thus the average delivery pressure $(Pps1+Pps2)/2$ of the two delivery ports **20a** and **20b** of the main pump **20** is calculated. This average delivery pressure $(Pps1+Pps2)/2$ of the main pump **20** is introduced to the table **55c**. Furthermore, the delivery pressure **Pps3** of the main pump **21** that is an output from the pressure sensor **41c** is introduced to the table **55c**.

(86) In the table **55c**, the same characteristic as the absorption torque characteristic (FIG. **6**) of the main pump **20** controlled by the torque control pistons **22f**, **22g**, and **22h** of the above-described regulator **22** is set. The average delivery pressure $(Pps1+Pps2)/2$ of the main pump **20** and the delivery pressure **Pps3** of the main pump **21** are introduced to the table **55c**. Reference to the average delivery pressure $(Pps1+Pps2)/2$ of the main pump **20** and the delivery pressure **Pps3** of the main pump **21** is made in the table **55c** and the capacity **q12** of the main pump **20** is calculated.

(87) The capacity **q12** of the main pump **20** calculated with the table **55c** is doubled by the gain **55d**.

(88) Furthermore, the target revolution speed **Nac** that is an input from the target revolution speed instruction dial **51** is introduced to the multiplying section **55e** together with the capacity **q12*2** calculated with the gain **55d** and a target flow rate **Q12ac** of the main pump **20** is calculated. The target flow rate **Q12ac** is the total of the delivery flow rates of two delivery ports **20a** and **20b** of the main pump **20**. This target flow rate **Q12ac** and the average delivery pressure $(Pps1+Pps2)/2$ of the main pump **20** calculated with the gain **55b** are introduced to the multiplying section **55f** and target power **Pw12ac** of the main pump **20** is calculated.

(89) Meanwhile, in the table **55h**, the same characteristic as the absorption torque characteristic (see FIG. **7**) of the above-described fixed displacement main pump **21** is set. The delivery pressure **Pps3** of the main pump **21** that is the output from the pressure sensor **41c** is introduced to the table **55h**. Reference to the delivery pressure **Pps3** of the main pump **21** is made in the table **55h** and the capacity **q3** of the main pump **21** is calculated. The table **55h** outputs the constant value **qmax3** as the capacity **q3** irrespective of the value of the delivery pressure **Pps3**.

(90) Since the capacity **qmax3** is constant, instead of calculating the capacity **qmax3** with use of the table **55h**, the constant capacity **qmax3** may be stored in a memory of the controller **55** and the capacity **qmax3** may be used.

(91) Furthermore, the target revolution speed **Nac** that is the input from the target revolution speed instruction dial **51** is introduced to the multiplying section **55i** together with the capacity **q3** calculated with the table **55h** and a target flow rate **Q3ac** of the main pump **21** is calculated. This target flow rate **Q3ac** and the delivery pressure **Pps3** of the main pump **21** that is the output from the pressure sensor **41c** are introduced to the multiplying section **55j** and target power **Pw3ac** of the main pump **21** is calculated.

(92) The target power **Pw12ac** calculated in the multiplying section **55f** and the target power **Pw3ac** calculated in the multiplying section **55j** are added in the adding section **55g** and total target power **Pw123ac** is calculated.

(93) Maximum allowable power **Pwmax** that is an output from the maximum allowable power

setting device **81** incorporated in the monitor **80** and the target power Pw_{123ac} calculated in the adding section **55g** are introduced to the minimum value selecting section **55k** and post-limiting power Pw_{reg} is calculated. The post-limiting power Pw_{reg} that is an output from the minimum value selecting section **55k** is introduced to the gain **55l** and the post-limiting power Pw_{reg} is multiplied by Pw_{12ac}/Pw_{123ac} , and thus post-limiting power Pw_{12reg} that can be used by the main pump **20** is calculated. Pw_{12ac}/Pw_{123ac} represents the ratio of the target power Pw_{12ac} of the main pump **20** of the variable displacement type calculated in the multiplying section **55f** to the total target power Pw_{123ac} of the main pump **20** of the variable displacement type and the main pump **21** of the fixed displacement type calculated in the adding section **55g**. In other words, Pw_{12ac}/Pw_{123ac} represents the power that can be consumed by the main pump **20** of the variable displacement type in the power limited to the maximum allowable power Pw_{max} .

(94) The post-limiting power Pw_{12reg} and the average delivery pressure $(Pps_1 + Pps_2)/2$ of the main pump **20** calculated with the gain **55b** are introduced to the dividing section **55m** and a post-limiting flow rate Q_{12reg} is calculated. The post-limiting flow rate Q_{12reg} and the capacity q_{12*2} calculated with the gain **55d** are introduced to the dividing section **55n** and a post-limiting revolution speed N_{reg} is calculated.

(95) The post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} that is the input from the target revolution speed instruction dial **51** are input to the minimum value selecting section **55o** and the smaller value of the post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} is selected as a command revolution speed N_d and is output to the inverter **60**.

(96) As above, in the present embodiment, the hydraulic drive system includes plural hydraulic pumps including the two main pumps **20** and **21** (first and second hydraulic pumps) and includes, as the pressure sensor, plural pressure sensors including the first pressure sensors **41a** and **41b** and the second pressure sensor **41c** that sense the delivery pressure of each of the two main pumps **20** and **21**. The controller **55** calculates the target power to be consumed by the two main pumps **20** and **21** (first and second hydraulic pumps) on the basis of the capacities of the two main pumps **20** and **21** (first and second hydraulic pumps), the delivery pressures of the two main pumps **20** and **21** sensed by the first pressure sensors **41a** and **41b** and the second pressure sensor **41c**, and the target revolution speed of the electric motor **1**.

(97) Furthermore, the main pump **20** (first hydraulic pump) is the variable displacement type and the main pump **21** (second hydraulic pump) is the fixed displacement type. The main pump **20** (first hydraulic pump) has the regulator **22** including the first torque control pistons **22f** and **22g** and the second torque control piston **22h** to which the delivery pressures of the main pump **20** and the delivery pressure of the main pump **21** (second hydraulic pump) are each introduced and that control the capacity of the main pump **20** in such a manner that the total of the absorption torque of the main pump **20** and the absorption torque of the main pump **21** does not exceed a predetermined value. In the table **55c** of the controller **55**, as the absorption torque characteristic of the main pump **20**, the absorption torque characteristic of the main pump **20** controlled by the first torque control pistons **22f** and **22g** and the second torque control piston **22h** is set.

(98) —Operation—

(99) Operation of the second embodiment will be described.

(100) When the operation lever of an optional operation lever device in the plural operation lever devices including the operation lever devices **24A** and **124B** (see FIG. 1) is operated, the corresponding directional control valve is switched and the hydraulic fluid is supplied to the corresponding actuator. At this time, the directional control valve is switched with a stroke according to the operation amount of the operation lever and the main pumps **20** and **21** driven by the electric motor **1** deliver a flow rate according to the revolution speed of the electric motor **1** and the absorption torque control of the torque control pistons **22f**, **22g**, and **22h** of the regulator **22**. The actuator is driven at a speed according to the operation amount of the operation lever.

(101) In the present embodiment, operation of the regulator **22** that carries out the absorption torque control and the directional control valves of the open center type is general and therefore details thereof are omitted.

(102) DC power supplied from the battery **70** or DC power supplied after being converted from AC power by the AC/DC converter **90** from the commercial power supply **92** through the connector **91** or the DC power of both is supplied to the inverter **60** that drives the electric motor **1** through the DC power supply line **65**.

(103) The maximum allowable power P_{wmax} set in advance is input to the controller **55** from the maximum allowable power setting device **81** incorporated in the monitor **80**.

(104) The outputs from the pressure sensors **41a**, **41b**, and **41c** as the pump delivery pressures P_{ps1} , P_{ps2} , and P_{ps3} and the output from the target revolution speed instruction dial **51** as the target revolution speed N_{ac} are each input to the controller **55**.

(105) Processing in the controller **55** will be described below with classification of cases.

(106) (a) Case in which the Target Power P_{w123ac} of the Main Pump **20** and the Main Pump **21** is the Same as or Lower than the Maximum Allowable Power P_{wmax} ($P_{w123ac} \leq P_{wmax}$)

(107) The maximum allowable power P_{wmax} and the target power P_{w123ac} are introduced to the minimum value selecting section **55k** and P_{w123ac} that is the minimum value is selected, and thus the post-limiting power P_{wreg} becomes $P_{wreg} = P_{w123ac}$.

(108) The post-limiting power P_{wreg} is multiplied by P_{w12ac}/P_{w123ac} at the gain **55l** and the post-limiting power P_{w12reg} that can be used by the main pump **20** becomes $P_{w12reg} = P_{wreg} (=P_{w123ac}) \times P_{w12ac}/P_{w123ac} = P_{w12ac}$.

(109) In the dividing section **55m**, $P_{w12reg}/(P_{ps1}+P_{ps2})/2$ is computed. At this time, $P_{w12reg} = P_{w12ac}$ holds in the case of $P_{w123ac} \leq P_{wmax}$. Therefore, the post-limiting flow rate Q_{12reg} becomes $Q_{12reg} = P_{w12reg}/(P_{ps1}+P_{ps2})/2 = P_{w12ac}/(P_{ps1}+P_{ps2})/2 = Q_{12ac}$.

(110) In the dividing section **55n**, $Q_{12reg}/(2 \times q_{12})$ is computed. At this time, $Q_{12reg} = Q_{12ac}$ holds as described above. Therefore, the post-limiting revolution speed N_{reg} becomes $N_{reg} = Q_{12reg}/(2 \times q_{12ac}) = Q_{12ac}/(2 \times q_{12}) = N_{ac}$.

(111) The post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} are input to the minimum value selecting section **55o** and the minimum value is selected. At this time, $N_{reg} = N_{ac}$ holds as described above. Therefore, the command revolution speed N_d output from the controller **55** to the inverter **60** becomes $N_d = N_{ac}$ without being limited by the minimum value selecting section **55o**.

(112) (b) Case in which the Target Power P_{w123ac} of the Main Pump **20** and the Main Pump **21** is Higher than the Maximum Allowable Power P_{wmax} ($P_{w123ac} > P_{wmax}$)

(113) The maximum allowable power P_{wmax} and the target power P_{w123ac} are introduced to the minimum value selecting section **55k**. In this case, the maximum allowable power P_{wmax} is selected as the minimum value and the post-limiting power P_{wreg} becomes $P_{wreg} = P_{wmax}$.

(114) The post-limiting power P_{wreg} is multiplied by P_{w12ac}/P_{w123ac} at the gain **55l** and $P_{w12reg} = P_{wreg} (=P_{wmax}) \times P_{w12ac}/P_{w123ac}$ is calculated regarding the post-limiting power P_{w12reg} that can be consumed by the main pump **20**.

(115) The post-limiting flow rate Q_{12reg} is calculated as

$Q_{12reg} = P_{wmax} \times P_{w12ac}/P_{w123ac}/(P_{ps1}+P_{ps2})/2$ by the dividing section **55m**. At this time, since originally a relation of $Q_{12ac} = P_{w12ac}/(P_{ps1}+P_{ps2})/2$ holds, a relation of $Q_{12reg}/Q_{12ac} = P_{wmax}/P_{w123ac} (<1)$ holds from these two expressions.

(116) Subsequently, the post-limiting revolution speed N_{reg} is calculated as

$N_{reg} = Q_{12reg}/(2 \times q_{12}) = Q_{12ac} \times (P_{wmax}/P_{w123ac})/(2 \times q_{12})$ by the dividing section **55n**. Also in this case, since originally a relation of $N_{ac} = Q_{12ac}/(2 \times q_{12})$ holds, a relation of $N_{reg}/N_{ac} = Q_{12reg}/Q_{12ac} = P_{wmax}/P_{w123ac} (<1)$ holds from these two expressions.

(117) The post-limiting revolution speed N_{reg} and the target revolution speed N_{ac} are input to the minimum value selecting section **55o**. At this time, since $N_{reg} < N_{ac}$ holds as described above, N_{reg}

that is the smaller value than the target revolution speed N_{ac} is selected as the command revolution speed N_d and is output from the controller **55** to the inverter **60**.

(118) —Effects—

(119) In the present embodiment, the following effects are obtained.

(120) 1. The controller **55** calculates the target power P_{w123ac} to be consumed by the main pumps **20** and **21** on the basis of the capacities q_{12} and q_3 of the main pumps **20** and **21**, the delivery pressures P_{ps1} , P_{ps2} and P_{ps3} of the main pumps **20** and **21** sensed by the pressure sensors **41a**, **41b**, and **41c**, and the target revolution speed N_{ac} of the electric motor **1** and outputs the command revolution speed N_d to the inverter **60** to limit the target revolution speed N_{ac} of the electric motor **1** in such a manner that the target power P_{w123ac} falls within the range of the maximum allowable power P_{wmax} . Therefore, similarly to the first embodiment, the power consumption of the electric motor **1** is surely limited to be equal to or lower than the maximum allowable power P_{wmax} . This prevents abnormal lowering of the voltage of the battery **70** that supplies electric power to the electric motor **1** and operation of the breaker of the commercial power supply **92** to the interruption position in operation of the electrically driven hydraulic work machine and can surely prevent a sudden stop of the front work implement **104**.

(121) Furthermore, the same effects as items **2** to **5** of the first embodiment are obtained, such as the effect that the operator of the electrically driven hydraulic work machine does not need to operate the target revolution speed instruction dial **51** of the electric motor **1** and therefore troublesomeness of operation can be eliminated.

(122) 2. The hydraulic drive system includes, as the hydraulic pump, plural hydraulic pumps including the two main pumps **20** and **21** (first and second hydraulic pumps) and includes, as the pressure sensor, plural pressure sensors including the first pressure sensors **41a** and **41b** and the second pressure sensor **41c** that sense the respective delivery pressures P_{ps1} , P_{ps2} and P_{ps3} of the two main pumps **20** and **21**. The controller **55** calculates the target power P_{w123ac} to be consumed by the two main pumps **20** and **21** (first and second hydraulic pumps) on the basis of the capacities q_{12} and q_3 of the two main pumps **20** and **21** (first and second hydraulic pumps), the delivery pressures P_{ps1} , P_{ps2} and P_{ps3} of the two main pumps **20** and **21** sensed by the first pressure sensors **41a** and **41b** and the second pressure sensor **41c**, and the target revolution speed N_{ac} of the electric motor **1**.

(123) Due to this, also when the hydraulic drive system includes the plural hydraulic pumps (main pumps **21** and **22**) as the hydraulic pump, the target power P_{w123ac} to be consumed by the plural hydraulic pumps (two main pumps **20** and **21**) can be calculated and the target revolution speed N_{ac} of the electric motor **1** can be limited in such a manner that the target power P_{w123ac} falls within the range of the maximum allowable power P_{wmax} .

(124) 3. The main pump **20** is the variable displacement type and the main pump **21** is the fixed displacement type. The regulator **22** of the main pump **20** (first hydraulic pump) includes the torque control pistons (first torque control pistons) **22f** and **22g** and the torque control piston (second torque control piston) **22h** to which the delivery pressures of the main pump **20** and the delivery pressure of the main pump **21** (second hydraulic pump) are each introduced and that control the capacity of the main pump **20** in such a manner that the total of the absorption torque of the main pump **20** and the absorption torque of the main pump **21** does not exceed a predetermined value, and carries out total torque control. Also in this case, the controller **55** can calculate the target power P_{w123ac} to be consumed by the two main pumps **20** and **21** and limit the target revolution speed N_{ac} of the electric motor **1** in such a manner that the target power P_{w123ac} falls within the range of the maximum allowable power P_{wmax} because the same absorption torque characteristic as the absorption torque characteristic of the main pump **20** is set in the table **55c** of the controller **55** and the same absorption torque characteristic as the absorption torque characteristic of the main pump **21** is set in the table **55h**.

DESCRIPTION OF REFERENCE CHARACTERS

(125) **1:** Electric motor **2:** Main pump of the variable displacement type (hydraulic pump) **3a to 3h:** Actuator **4:** Control valve block **5:** Hydraulic fluid supply line **5a, 5b, 5c:** Hydraulic fluid supply line **6a to 6c:** Directional control valve **7a to 7c:** Pressure compensating valve **8a to 8c:** Check valve **9a to 9c:** Shuttle valve **11:** Differential pressure reducing valve **12, 22:** Regulator **12d:** Torque control piston **12e, 22e:** Spring **12c:** Flow rate control piston **12b:** LS valve **13:** Prime mover revolution speed sensing valve **14:** Main relief valve **14a, 14b, 14c:** Main relief valve **15:** Unloading valve **15a, 15c, 15d:** Pressure receiving part **15b:** Spring **20:** Main pump of the variable displacement type (first hydraulic pump) **21:** Main pump of the fixed displacement type (second hydraulic pump) **22f, 22g, 22h:** Torque control piston **30:** Pilot pump **31a, 31b, 31c:** Pilot pressure supply line **24:** Gate lock lever **32:** Pilot relief valve **40:** Control valve block **41:** Pressure sensor **41a, 41b, 41c:** Pressure sensor **50, 55:** Controller **51:** Target revolution speed instruction dial **60:** Inverter **65:** DC power supply line **70:** Battery **80:** Monitor **81:** Maximum allowable power setting device **90:** AC/DC converter **91:** Connector **92:** Commercial power supply **100:** Selector valve

Claims

1. An electrically driven hydraulic work machine comprising an electric motor, a hydraulic pump driven by the electric motor, an inverter for controlling a revolution speed of the electric motor, and a controller that outputs a command revolution speed for controlling the revolution speed of the electric motor on a basis of a target revolution speed of the electric motor, and drives the hydraulic pump to carry out work, wherein the electrically driven hydraulic work machine comprises: a maximum allowable power setting device that sets maximum allowable power allowed to be consumed by the electric motor, a pressure sensor that senses a delivery pressure of the hydraulic pump, and a target revolution speed instruction device that makes an instruction of the target revolution speed of the electric motor, and the controller is configured to calculate target power to be consumed by the hydraulic pump on a basis of a capacity of the hydraulic pump, the delivery pressure of the hydraulic pump sensed by the pressure sensor, and the target revolution speed of the electric motor instructed by the target revolution speed instruction device, calculate a first target revolution speed of the electric motor on a basis of a lower one of the target power and the maximum allowable power set by the maximum allowable power setting device, select, as a second target revolution speed, a lower one of the calculated first target revolution speed and the target revolution speed of the electric motor instructed by the target revolution speed instruction device, and outputting the selected second target revolution speed to the inverter as the command revolution speed, wherein the converter controls the revolution speed of the electric motor on a basis of the command target revolution speed in such a manner that the target power falls within a range of the maximum allowable power, wherein the hydraulic pump includes a plurality of hydraulic pumps including a first hydraulic pump and a second hydraulic pump, the pressure sensor includes a plurality of pressure sensors including a first pressure sensor that senses a delivery pressure of the first hydraulic pump and a second pressure sensor that senses a delivery pressure of the second hydraulic pump, and the controller is configured to calculate power to be consumed by the first and second hydraulic pumps as the target power on a basis of capacities of the first and second hydraulic pumps, the delivery pressures of the first and second hydraulic pumps sensed by the first and second pressure sensors, and the target revolution speed of the electric motor, and wherein the first hydraulic pump is a variable displacement type and the second hydraulic pump is a fixed displacement type, the first hydraulic pump has a regulator including first and second torque control pistons to which the delivery pressure of the first hydraulic pump and the delivery pressure of the second hydraulic pump are introduced respectively and that control the capacity of the first hydraulic pump in such a manner that a total of absorption torque of the first hydraulic pump and absorption torque of the second hydraulic pump does not exceed a predetermined value, the controller includes a first table in which the same characteristic as the absorption torque

characteristic of the first hydraulic pump controlled by the first and second torque control pistons is set, and a second table in which the same characteristic as the absorption torque characteristic of the second hydraulic pump is set, the controller is further configured to: refer to the delivery pressure of the first hydraulic pump sensed by the first pressure sensor in the first table to calculate the capacity of the first hydraulic pump, and calculate the first target power by using the calculated capacity of the first hydraulic pump, the target revolution speed of the electric motor, and the delivery pressure of the first hydraulic pump sensed by the first pressure sensor, refer to the delivery pressure of the second hydraulic pump sensed by the second pressure sensor in the second table to calculate the capacity of the second hydraulic pump, and calculate the second target power by using the calculated capacity of the second hydraulic pump, the target revolution speed of the electric motor, and the delivery pressure of the second hydraulic pump sensed by the second pressure sensor, and calculate the target power by adding the first target power and the second target power.

2. The electrically driven hydraulic work machine according to claim 1 wherein the controller is further configured to: calculate the first target revolution speed of the electric motor by dividing a lower of the target power and the maximum allowable power set by the maximum allowable power setting device by the delivery pressure of the hydraulic pump and the capacity of the hydraulic pump.

3. The electrically driven hydraulic work machine according to claim 1, further comprising a power supply that supplies electric power to the electric motor, wherein a plurality of values are stored in the maximum allowable power setting device according to kinds of the power supply as the maximum allowable power.
